

Javier González Sempere

Mathematical Engineering and Physics Student — Seeking Data / Consulting / Operations / Technology Internships

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LinkedIn | GitHub | Portfolio

Profile

Mathematical Engineering and Physics student with hands-on experience in data analysis, mathematical modelling, simulation, optimisation and instrumentation. Develops end-to-end projects using Python and physical systems, combining technical rigour, measurable results and clear communication.

Education

Double Bachelor's Degree in Mathematical Engineering and Physics — Universidad CEU San Pablo 2025–2030 (Expected)

Selected Research and Projects

Airport Queue Optimisation System

Computer Vision, Queueing Theory and Operations

- Developed a decision-support system integrating passenger counts from **YOLO/OpenCV**, M/M/c queueing models, simulation and a Streamlit dashboard to estimate congestion, waiting times and bottleneck risk.
- Compared operational scenarios across **5 airport zones** and reduced simulated cumulative waiting time from **19.09 to 5.32 minutes (72.1% reduction)**.

Railway Structural Health Monitoring

IoT, Signal Processing and Predictive Maintenance

- Integrated acceleration, temperature and piezoelectric-response data collected with an ESP32 into a team-built railway structural monitoring prototype.
- Built a Python pipeline for data cleaning, filtering, normalisation and feature extraction, together with an interpretable damage-scoring function and a Streamlit dashboard for condition monitoring and alerts.

Supermarket Operations Optimisation

Analytics, Forecasting and Operations Research

- Conducted an end-to-end analysis of **2.54 million records**, combining exploratory data analysis, queueing theory, demand forecasting, product co-occurrence analysis and thermal simulation.
- Achieved a weekly demand-forecasting **MAE of 0.77** and evaluated operational policies for checkout capacity, inventory replenishment and energy management.

Materials Physics Research

Manuscript Under Review at SURE-J

- Designed a low-cost experimental setup to compare the transient thermal response of a conventional hollow brick and a paraffin/graphite-modified brick.
- Measured a **503-second delay** in reaching $\Delta T = 2^\circ\text{C}$, processed the experimental signals, and contributed to reduced-order modelling and manuscript preparation.

Formula 1 Pit-Stop Strategy

Physics-Based Simulation and Historical Validation

- Developed an interpretable race simulator modelling tyre degradation, pit-lane time loss and strategy sensitivity to the number of pit stops.
- Validated model trends against **10,666 cleaned historical pit-stop records**, analysing the gap between the theoretical optimum and strategies observed in competition.

Leadership and Technical Communication

Applied Technical Projects Club — Founder and Director

- Founded and coordinate an interdisciplinary university initiative focused on applied projects in science, data, simulation, optimisation and physical systems.

Technical Portfolio and Writing

- Maintain a public technical portfolio documenting project methodology, results, limitations and lessons learned, together with essays on science, technology and decision-making.

Competitions

PLANCKS National 2026: ranked **21st out of 162 teams** in the Spanish national round of the international theoretical physics competition.

imagin Hackathon: collaborated in a team to develop a sustainability-focused proposal addressing a real-world challenge through innovation and entrepreneurship.

Skills and Languages

Programming and Data: Python, Julia, pandas, NumPy, SciPy, scikit-learn, matplotlib, Plotly.

Tools: Git, GitHub, Jupyter, LaTeX, Streamlit.

Computer Vision and IoT: OpenCV, YOLO, ESP32, MPU6050, MAX6675, piezoelectric sensors and data acquisition.

Methods: mathematical modelling, simulation, queueing theory, forecasting, signal processing and experimental research.

Languages: English C1, Valencian C1, German B1, French B1.